Prospective low-prevalence evaluation

10,000 new URLs • 5.00% phishing • frozen models and thresholds

Condition	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC	FPR	FNR
Weighted HYB-MS real-only	98.00%	76.88%	85.80%	81.10%	97.22%	87.24%	1.36%	14.20%
Weighted HYB-MS +10% GAN	98.45%	81.89%	88.60%	85.11%	97.30%	91.11%	1.03%	11.40%

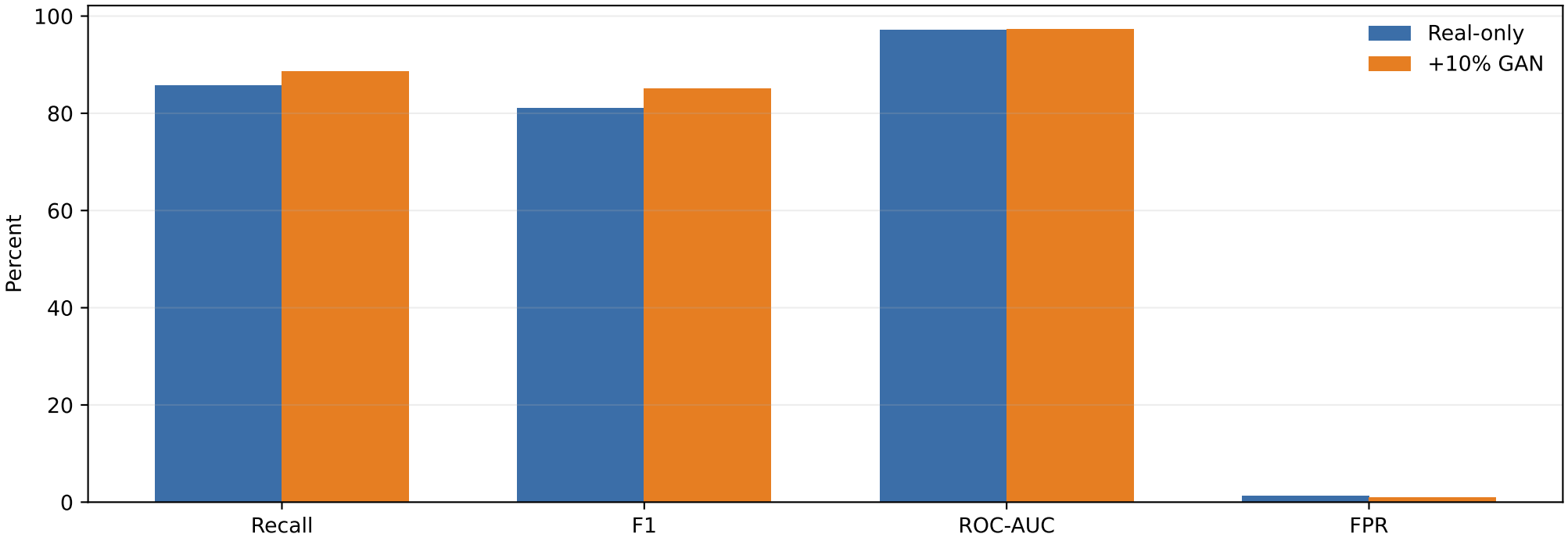
AUGMENTATION BENEFIT SUPPORTED

The +10% model improved recall, F1 and ROC-AUC without increasing FPR on this new cohort.

No training, calibration, checkpoint choice or threshold was changed after this dataset was frozen.

Dataset SHA-256: b331bd254ec9141e72b24489f0a0bba89951c4d410a211caf9c2d8f7924501dd

What the result means in browser-like traffic



Real-only false alerts per 10,000 visits: 129.0

+10% GAN false alerts per 10,000 visits: 98.0

This is an identity-disjoint, proxy-labelled evaluation. It is useful evidence, but it is still not a live production deployment test.